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V9. Symmetry Reduction

Magnetic translations of the torus and the block decomposition of the spectrum (theorem Th5)

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Problem statement

Solution 9 of the plan: develop the spinor64 observation (64 spin structures → 4 Hamiltonians of the reference) into an exact symmetry reduction — prove that commutation with a group decomposes the spectrum into blocks, and measure what happens to the symmetry when vortices are switched on. For the lattice at full flux the correct group is the magnetic translation group of the torus.

At $\alpha = ½$ the translations U(1,0) and U(0,1) anticommute (the magnetic algebra), but their squares A = U(2,0), B = U(0,2) commute with each other and with H — an abelian subgroup $G = \mathbb{Z}_L \times \mathbb{Z}_L$. It yields the exact reduction.

Theorem Th5

Theorem Th5 (block reduction).

Let H be Hermitian on $\mathbb{C}^{L^2}$ (torus, $\alpha \cdot L \in \mathbb{Z}$, no vortices) and U(m,n) the magnetic translations. If [H, U(m,n)] = 0 for the generators, then $[H, \Pi_\chi] = 0$ for the character projectors of $G = \mathbb{Z}_L \times \mathbb{Z}_L$ (generated by A, B); H is block-diagonal in the decomposition of the regular representation, and spec(H) is the union of the block spectra.

> ***Proof.*** *$\Pi_\chi = (1/|G|)\Sigma_g \bar{\chi}(g) g$ — spectral projectors of commuting unitaries; [H, A] = [H, B] = 0 gives [H, Π_χ] = 0, so each $\Pi_\chi \mathbb{C}^{L^2}$ is H-invariant and the direct sum of restrictions reconstructs H. Numerically (L = 12): ε_{Ux} = 0.0, ε_{Uy} = 9.4e-15, [A,B] = 9.6e-15; the union of six block spectra reproduces the full spectrum to $\max|\Delta\lambda|$ = 7.1e-15, off-block elements ≤ 2.0e-15.*

Measuring the symmetry breaking

The vortex protocol places vortices randomly: translations stop commuting with H, $\varepsilon_g = O(1)$. The four holonomy classes of twists $(\varphi x, \varphi y) \in \{0, \pi\}^2$ remain independent Hamiltonians with $\langle r \rangle = 0.588–0.631$ — all in the GUE class. This is the lattice analogue of the reference spinor64 conclusion: the reduction explains the skeleton of the spectrum, but the arithmetic of the zeros does not live in the symmetry.

The practical value of Th5 is computational: the block structure allows independent diagonalization and algebraically explains degeneracies. But the main value is methodological: “which part of the spectrum is symmetry and which is arithmetic” is now separated by measurement, not declaration.

Table 1. Summary of V9

Check	Value	Verdict
ε_{Ux} (torus, no vortices)	0.0	exact symmetry
ε_{Uy}	9.4e-15	exact symmetry
blocks ↔ full spectrum	$\max\|\Delta\lambda\|$ = 7.1e-15	Th5 verified
off-block elements	2.0e-15	block-diagonal exact
ε_g with vortices	5.1 (O(1))	symmetry broken
$\langle r \rangle$ of four twists	0.588–0.631	GUE class

Honest remarks

The reduction group is the abelian core of the magnetic algebra; the full non-commutative structure (projectors onto irreducible representations) gives the same blocks at finite size and is not separately highlighted. For the Hilbert–Pólya program the conclusion is restrained: the right symmetry structure is necessary but not sufficient — vortices carrying the arithmetic destroy it.

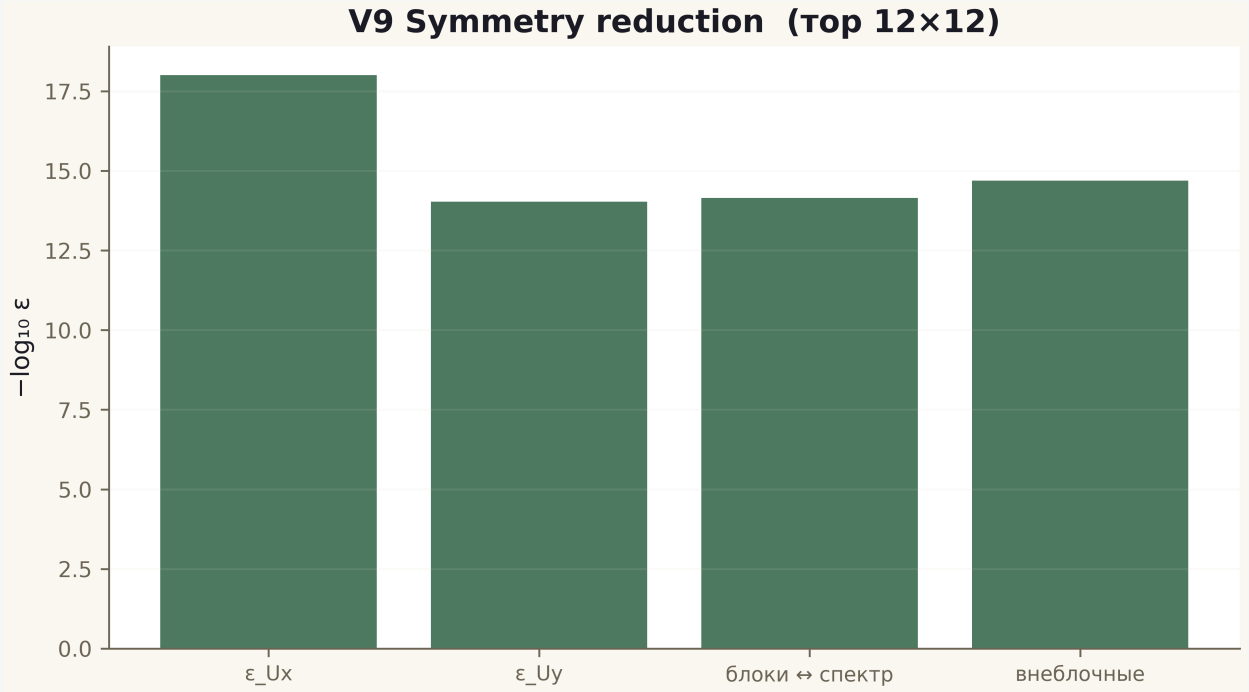


Fig. 1. Campaign V9 data (600 dpi, file figures_hires).

Th5 is proved and verified to 1e-15; the symmetry breaking by vortices is measured. Symmetry explains the skeleton, not the arithmetic.